Converting Mass, Area and Volume

Question Bank 5.08 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, convert with the ratio given.

1. $1\,\text{lb} = 0.454\,\text{kg}$. 20 lb to kilograms.

9.08 kg

(20 ÷ 2.204 = 9.07 by the other ratio.)

2. $1\,\text{kg} = 2.204\,\text{lb}$. 15 kg to pounds.

33.06 lb

3. $1\,\text{gal} = 3.785\,\text{L}$. 8 gal to liters.

30.28 L

4. $1\,\text{L} = 0.264\,\text{gal}$. 25 L to gallons.

6.6 gal

For questions 5 through 7, say how many times the ratio is used.

5. $1\,\text{yd} = 3\,\text{ft}$. $4\,\text{yd}^2$ to square feet.

36 ft²

4 × 9: twice.

6. $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$. 32 fl oz to liters.

0.96 L

4 c × 0.24.

7. $1\,\text{ft} = 12\,\text{in}$. $3\,\text{ft}^3$ to cubic inches.

5,184 in³

3 × 1,728: three times.

Choose one.

8. $1\,\text{ft} = 12\,\text{in}$. A rug is $6\,\text{ft}^2$ in area. How many square inches is that?

- ● **A.** $864\,\text{in}^2$
- ○ B. $72\,\text{in}^2$
- ○ C. $144\,\text{in}^2$
- ○ D. $150\,\text{in}^2$

A

6 × 144.